

Temporal Analysis of Deforestation in Central Borneo Through NDVI-Based Remote Sensing

Multi-Temporal Vegetation Change Assessment Using Sentinel-2, Google Earth Engine,
R, and QGIS (2018–2024)

Author: Israel Vallejo Muñoz

Marine Biologist (MSc) & Environmental Data Analyst

Tools: Google Earth Engine · Sentinel-2 · MODIS · QGIS · R

Date: June 2025 | GIS & Remote Sensing Portfolio



1. Introduction

Forests cover approximately one-third of the Earth's land surface and provide essential ecosystem services, including regulation of the hydrological cycle, soil stabilization, climate mitigation, and biodiversity preservation (Sheram, 1993). Despite their importance, extensive deforestation remains a global environmental crisis, with at least 145 countries currently engaged in wood production (Chakravarty et al., 2012). For decades, the relentless destruction of forests has continued, and only recently have we begun to grasp its full magnitude.

Deforestation is defined as the conversion of forested land to permanent non-forested uses such as agriculture, urban development, or grazing, resulting in biodiversity loss and contributing to the greenhouse effect (van Kooten & Bulte, 2000; Angelsen et al., 1999). Forests act as major carbon sinks, storing more than 360 Pg of the 450–650 Pg of global vegetative carbon (Pan et al., 2013; IPCC, 2013). When soil is included, this figure exceeds 800 PgC, nearly equivalent to the total carbon in the atmosphere. Consequently, forest loss not only releases large quantities of carbon but also reduces future carbon sequestration potential (Houghton & Nassikas, 2018).

Tropical forests play a critical role in global and regional climate regulation due to their impact on hydrological and atmospheric dynamics (Kumagai et al., 2016). In Southeast Asia, deforestation rates are among the highest in the world (Canadell et al., 2007), with Borneo standing out as a deforestation hotspot over recent decades (Bryan et al., 2013; Gaveau et al., 2013, 2014). These changes have even been linked to reduced annual precipitation on the island (Kumagai et al., 2013).

Borneo, the third-largest island on Earth, is shared by Indonesia, Malaysia, and the Sultanate of Brunei. It harbors one of the most biodiverse and ecologically unique tropical forests in the world. Much of its lowland terrain is composed of peatland forests, which are naturally fire-resistant (Siegert et al., 2009; Page & Hooijer, 2016). However, increased fire frequency has been observed due to deforestation-driven drying, anthropogenic ignition, and land drainage (Herawati & Santoso, 2011; Wösten et al., 2008; Hoscilo et al., 2011). Between 2001 and 2016, Borneo experienced one of the highest rates of tropical peatland forest loss globally (Austin et al., 2019).

Deforestation is a major risk factor for wildfires and leads to measurable increases in local temperatures, compounding climate and fire vulnerabilities (Davies-Barnard et al., 2023). Beyond environmental concerns, Borneo's biodiversity is at risk: the island is home to iconic and endangered species such as the orangutan, the proboscis monkey, and countless other endemic species whose habitats are vanishing rapidly.

2. Objectives

Given this alarming context, this report aims to visualize and assess deforestation trends in a selected region of Central Borneo between 2018, 2020, and 2024. Through remote sensing and NDVI analysis, the objective is to evaluate the magnitude and direction of forest cover change, and to determine whether deforestation has progressed, stabilized, or shown signs of reversal.

- Calculate NDVI for 2018, 2020, and 2024 using Sentinel-2 imagery via Google Earth Engine.
- Classify vegetation change into Loss, No Change, and Gain categories for two time intervals.
- Analyze NDVI change by land cover type using MODIS classification and R statistical tools.
- Map the spatial distribution of vegetation change using QGIS thematic cartography.
- Quantify the acceleration of deforestation between the short-term and long-term periods.

3. Methods

To assess the spatial and temporal changes in vegetation cover across a region of Central Borneo, a multi-platform geospatial workflow was implemented combining satellite remote sensing, land cover classification, and statistical modeling. The study area is defined by the following coordinates: North -2.3° , South -2.9° , West 113.5° , East 114.3° (WGS 84, EPSG:4326).

3.1 NDVI Calculation (Google Earth Engine)

Sentinel-2 Level-2A surface reflectance imagery was used to generate cloud-free median composites for June–August in 2018, 2020, and 2024. NDVI was calculated as $(B8 - B4) / (B8 + B4)$. Two NDVI difference rasters were generated: $NDVI_{2024} - NDVI_{2018}$ (long-term, 6-year period) and $NDVI_{2024} - NDVI_{2020}$ (short-term, 4-year period). These rasters provided the basis for evaluating vegetation dynamics over time.

3.2 Land Cover Classification and Point Sampling

MODIS Land Cover Type 1 (MCD12Q1, year 2020) was used to classify land cover types within the study area. A random sample of 87 points was extracted from the NDVI difference raster using GEE. For each point, NDVI values for 2018, 2020, and 2024 were recorded, along with the NDVI difference (2024 – 2020) and land cover type (2020). The sample was exported as a .csv file for analysis in R.

3.3 Statistical and Graphical Analysis (R)

In R, the sampled data were processed to recode MODIS land cover classes into descriptive labels (Deciduous Forest, Savannas, Grasslands), calculate mean NDVI per land cover type for 2020 and 2024, and visualize the results through a boxplot of NDVI change by class and a scatterplot comparing NDVI 2020 vs. 2024.

3.4 Raster Classification and Mapping (QGIS)

The NDVI difference rasters were imported into QGIS and classified into three categories based on the following thresholds: Loss (NDVI difference < -0.1), No Change ($-0.1 \leq NDVI \leq 0.1$), and Gain (NDVI difference > 0.1). Thematic maps were produced for both time intervals, and raster statistics were used to compute the proportion of area within each change category.

Component	Source / Tool	Details
NDVI imagery	Sentinel-2 L2A (GEE)	Bands B8/B4 · Cloud-free median · June–Aug
Land cover	MODIS MCD12Q1 (2020)	Type 1 classification · 500 m resolution
Point sampling	Google Earth Engine	87 random points · NDVI + land cover
Statistical analysis	R	Boxplot, scatterplot, mean NDVI by class
Change mapping	QGIS	3-class reclassification · thematic maps
Regional mean NDVI	GEE reduceRegion()	Whole-area mean for 2020 and 2024

Table 1. Summary of data sources and analytical tools used in this study.

4. Results

4.1 NDVI Classification and Area Change (QGIS)

The NDVI difference rasters were reclassified into Loss, No Change, and Gain. The results revealed widespread degradation across both time periods, with the most significant losses recorded between 2020 and 2024.

Time Period	Loss (%)	No Change (%)	Gain (%)
2018 to 2024 (6-year)	87.58	8.65	3.76
2020 to 2024 (4-year)	96.34	2.70	0.96

Table 2. Proportion of area classified as vegetation Loss, No Change, and Gain for the two study periods.

These values indicate a clear acceleration in vegetation loss during the most recent four-year period. While nearly 9% of the area remained stable between 2018 and 2024, less than 3% showed stability in the 2020–2024 interval, and areas with vegetation gain fell from 3.76% to under 1%.

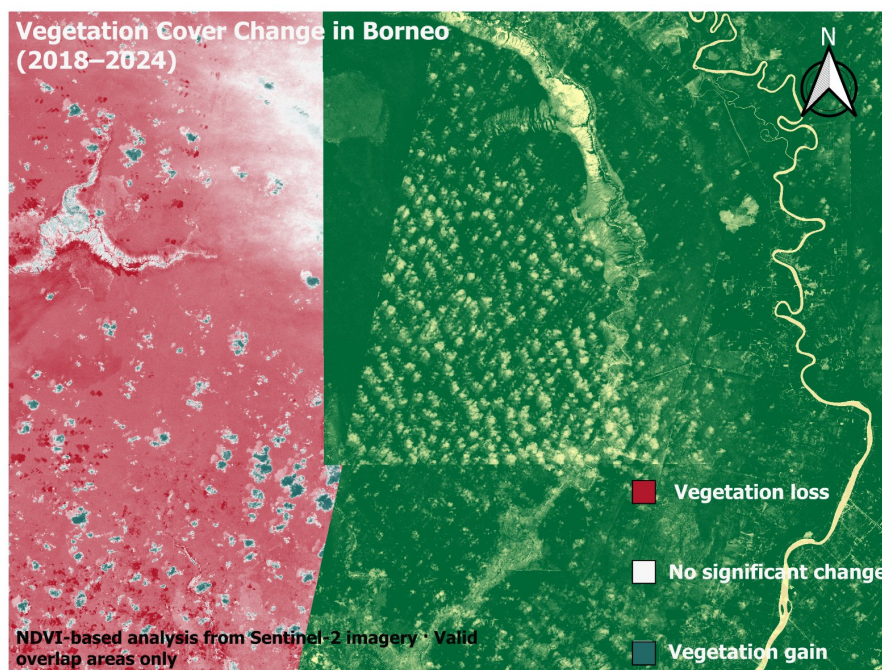


Figure 1. Vegetation Cover Change in Central Borneo (2018–2024). Left panel (red) shows vegetation loss; right panel (green) shows a 2024 Sentinel-2 true-colour composite for comparison. Source: Sentinel-2 / GEE. Analysis: Israel Vallejo Muñoz.

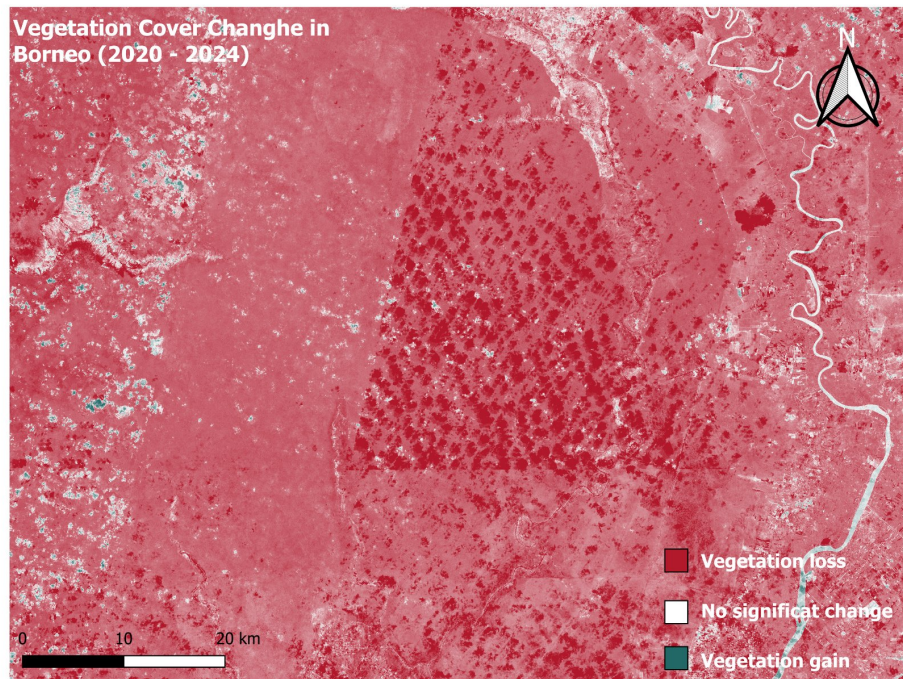


Figure 2. Vegetation Cover Change in Central Borneo (2020–2024). The near-total dominance of red tones confirms that 96.3% of the study area experienced vegetation loss over this four-year period. Source: Sentinel-2 / GEE. Analysis: Israel Vallejo Muñoz.

4.2 NDVI Change by Land Cover Type (R)

The 87 sampled points were analyzed by MODIS 2020 land cover class. All three classes recorded substantial NDVI declines between 2020 and 2024, with Savannas showing the largest average drop (−0.389), closely followed by Deciduous Forest (−0.355). Although Grasslands were underrepresented in the sample, they also showed a meaningful reduction (−0.237), indicating that vegetation degradation extends across all ecosystem types in the study area (Table 3).

Land Cover Type	NDVI 2020	NDVI 2024	Δ NDVI	n
Deciduous Forest	0.798	0.444	−0.355	70
Savannas	0.805	0.415	−0.389	16
Grasslands	0.713	0.476	−0.237	1

Table 3. Mean NDVI by land cover type for 2020 and 2024, with calculated NDVI change and sample size per class.

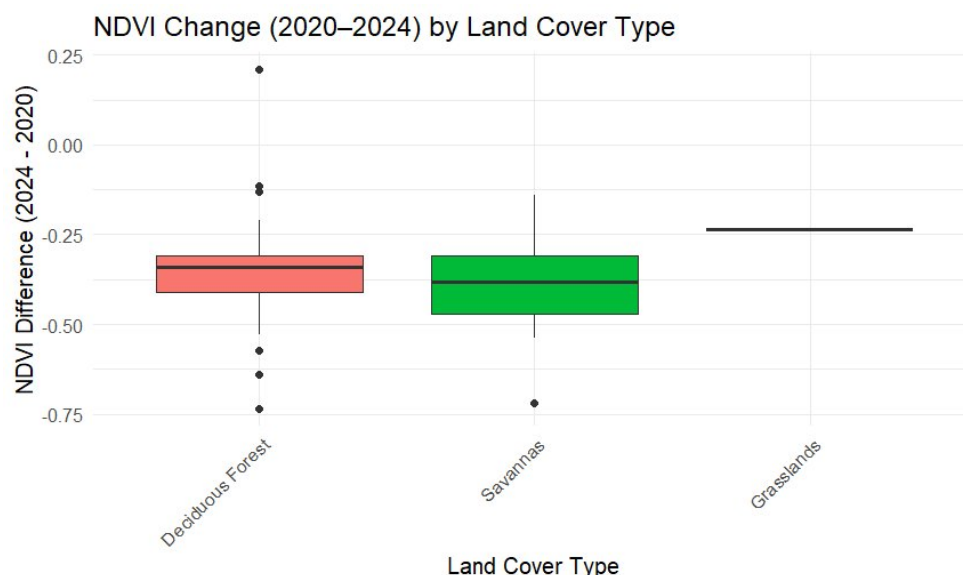


Figure 3. Boxplot of NDVI change (2020–2024) by MODIS land cover class. Both Deciduous Forest and Savannas show strong negative median values, with considerable spread. Source: MODIS / GEE / R. Analysis: Israel Vallejo Muñoz.

4.3 NDVI Scatterplot: 2020 vs. 2024

The scatterplot (Figure 4) comparing NDVI values in 2020 and 2024 for all sampled points shows that the majority of observations fall clearly below the 1:1 reference line, confirming a consistent and widespread decline in vegetation productivity. Points from all three land cover classes follow this downward pattern, with no cluster showing values systematically above the reference line.

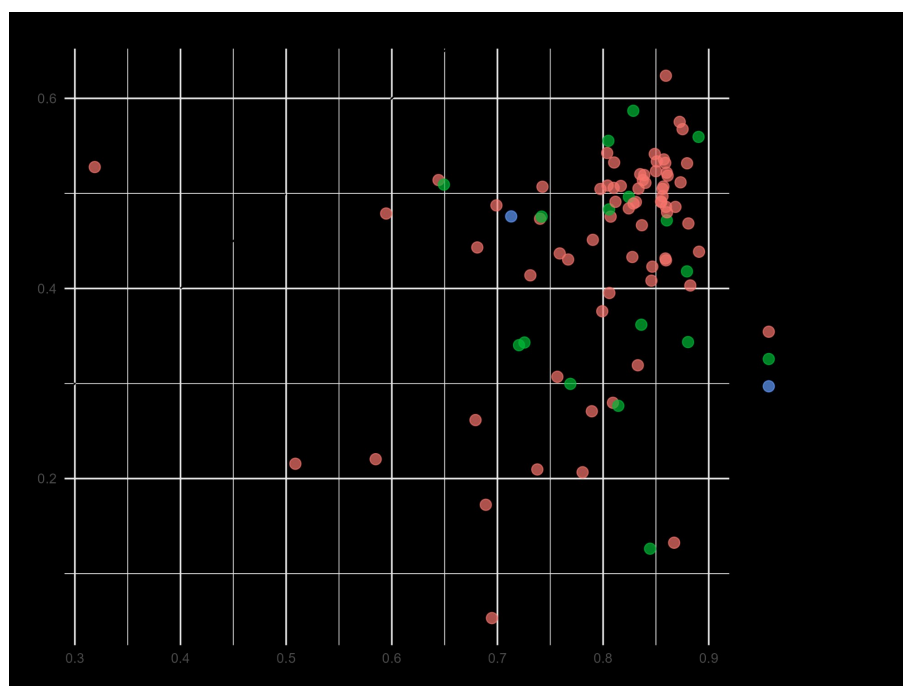


Figure 4. Scatterplot of NDVI 2020 vs. NDVI 2024 for all 87 sampled points, colored by MODIS land cover type. Points below the 1:1 diagonal indicate a decline in NDVI. Source: GEE / R. Analysis: Israel Vallejo Muñoz.

4.4 Histogram of NDVI Difference

The histogram of NDVI difference values (Figure 5) shows a strongly left-skewed distribution, with the vast majority of pixel values falling between -0.5 and -0.2 . This confirms that negative NDVI change is the dominant

signal across the study area, with very few pixels recording neutral or positive values.

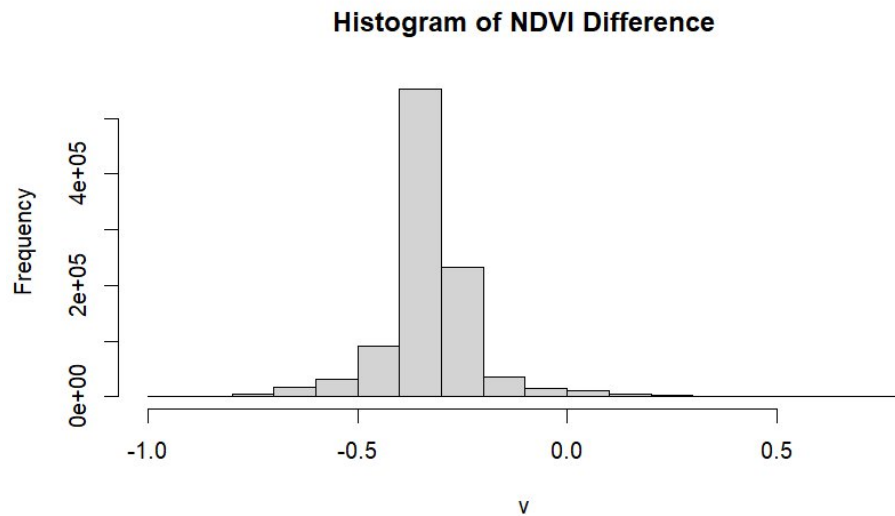


Figure 5. Histogram of NDVI difference ($NDVI_{2024} - NDVI_{2020}$) for all pixels in the study area. The strong left skew confirms that vegetation loss dominates the signal. Source: GEE / R. Analysis: Israel Vallejo Muñoz.

4.5 Regional Mean NDVI (GEE Output)

A region-wide mean NDVI was calculated directly in Google Earth Engine using the `reduceRegion()` function. The results confirm the degradation trend observed in all other analyses:

- Mean NDVI 2020: 0.7799
- Mean NDVI 2024: 0.4492
- Difference: -0.3307

This drop of 0.33 NDVI units in just four years is substantial and consistent with both the QGIS classification and the R-based point analysis.

5. Discussion

The results of this study clearly indicate a rapid and accelerating loss of vegetation cover in the selected region of Central Borneo between 2018 and 2024. NDVI-based analysis revealed that the proportion of area experiencing significant vegetation loss increased from 87.6% (2018–2024) to 96.3% (2020–2024), with a concurrent decrease in areas showing no change or vegetation gain. This dramatic trend suggests an intensification of deforestation and ecosystem degradation in recent years.

The boxplot and scatterplot analyses reinforce this finding across all three land cover types. Deciduous Forests and Savannas showed mean NDVI declines of approximately -0.36 and -0.39 , respectively. Although Grasslands were underrepresented in the sample, they too showed a noticeable NDVI decline, suggesting that degradation extends beyond dense forest areas to more open ecosystems.

Key Finding: The regional NDVI dropped from 0.78 to 0.45 in just four years (2020–2024), a decline of 0.33 units. This is consistent across all analytical platforms (GEE, R, QGIS) and across all land cover types, confirming an accelerating and near-total loss of vegetation cover in the study area, with 96.3% of pixels classified as vegetation loss.

These findings are consistent with previous reports highlighting Borneo as a deforestation hotspot (Bryan et al., 2013; Gaveau et al., 2014). The sharp reduction in vegetation cover aligns with documented land-use transitions, particularly those driven by oil palm expansion, illegal logging, and peatland drainage (Austin et al., 2019; Herawati & Santoso, 2011). The loss of forest cover not only impacts biodiversity but also exacerbates fire risk and regional climate alteration. Studies have shown that such land-use change reduces rainfall, dries out peatlands, and makes them more susceptible to ignition (Kumagai et al., 2013; Page & Hooijer, 2016).

The NDVI trend also mirrors larger-scale patterns of reduced forest resilience under anthropogenic pressure and climate stress (Davies-Barnard et al., 2023). The regional NDVI drop of -0.33 in just four years reflects a shift severe enough to influence carbon dynamics, evapotranspiration, and local hydrology. As Pan et al. (2013) and Houghton & Nassikas (2018) pointed out, forests play a central role in global carbon balance, and Borneo's continued loss could undermine both local and global mitigation efforts.

It is worth noting that the NDVI method, while widely used, has some limitations. It does not distinguish between tree species or forest structure, and can be influenced by atmospheric conditions or understory vegetation. However, the use of multi-year Sentinel-2 composites, consistent classification thresholds, and cross-platform validation strengthens the robustness of this study's conclusions.

6. Conclusions

This study reveals a concerning trend of intensified deforestation and vegetation degradation in Central Borneo between 2018 and 2024. Through NDVI analysis across multiple platforms, the results consistently demonstrate that:

- Vegetation loss has accelerated significantly, with 96.3% of the study area affected in the 2020–2024 period.
- Forested and non-forested ecosystems alike are experiencing severe degradation.
- The decline in NDVI (-0.33 units) is consistent across GEE, R, and QGIS analyses.
- These changes align with known environmental drivers in the region, including land-use conversion, oil palm expansion, peatland drainage, and fire susceptibility.

Given the ecological and climatic importance of Borneo's forests, the continued loss of vegetation poses a threat not only to local biodiversity (including orangutans and proboscis monkeys) but also to broader climate stability. Immediate and targeted conservation policies are urgently needed to curb forest loss, protect peatland ecosystems, and ensure long-term sustainability.

Future work should integrate additional variables such as precipitation records, fire hotspot data, elevation, and human activity layers to better understand the spatial drivers of deforestation. Higher-resolution land cover data and field validation would also strengthen the accuracy of remote sensing assessments.

7. References

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